

Building Age from Space

A transferable building-age classifier from 42 years of Landsat imagery — final report

Potato Team
Sara S. Ali & Mostafa Mohammed

Hack4Dev · November 2026

Contents

1 Summary

2 The task

3 Data

4 Planning and evaluation protocol

5 Preprocessing: from raw scenes to model inputs

6 Model development, step by step

7 The final backbone

8 The transfer method: modelCreate

9 Results

10 Discussion: what worked, what did not, limitations

11 Reproducibility

Appendix A Architecture and loss sweep

Appendix B Per-fold Madrid scores

1 Summary

We built a method that maps the construction period of every building pixel in a city from Landsat satellite time series (1984–2025), learning only from Madrid and adapting to a new city from a small number of labelled pixels. A **backbone** of 5 one-dimensional convolutional networks (TempCNN) reads each pixel's 42-year history of 32 measurements and turns it into an embedding in which buildings of the same period lie close together (supervised contrastive learning). The function **modelCreate** takes n labelled pixels per period from a new city and returns that city's classifier: with n ≤ 20 it classifies by distance to the labelled pixels' class prototypes; with n > 20 it recalibrates the network's normalisation on the new city, fine-tunes the top layers when cross-validation supports it, and applies a confidence rule in which uncertain pixels follow their spatial neighbours.

On Madrid, 10-fold spatial-block cross-validation gives macro-F1 **0.649 ± 0.038** (accuracy 67.0%). On Amsterdam, over 10 random draws per sample size, macro-F1 is **0.651** (n = 5), **0.685** (n = 25), **0.703** (n = 50), **0.738** (n = 100), **0.760** (n = 200); average over the five sample sizes 0.707, peak 0.760. Every pixel also receives a confidence level.

	Labels per period	Method	Macro-F1 (mean ± sd)	Accuracy
Madrid	10-fold spatial CV	backbone (5 networks)	0.649 ± 0.038	0.670
Amsterdam	n = 5	bare backbone	0.651 ± 0.017	0.659
Amsterdam	n = 25	full pipeline	0.685 ± 0.028	0.691
Amsterdam	n = 50	full pipeline	0.703 ± 0.022	0.706
Amsterdam	n = 100	full pipeline	0.738 ± 0.007	0.740
Amsterdam	n = 200	full pipeline	0.760 ± 0.006	0.762

Table 1. Headline results. Amsterdam: 10 random draws of n labelled pixels per period (seed 0); scored on every other labelled pixel of the dataset.

2 The task

The Hack4Dev challenge asks for a *method that creates classifiers*, not a single classifier: Stage 1 trains a transferable model on Madrid; Stage 2 adapts it to Amsterdam with 5, 25, 50, 100 and 200 labelled pixels per class. Required outputs are the Madrid cross-validated macro-F1 (mean and standard deviation) and confusion-matrix proportions, and Amsterdam macro-F1 (mean and standard deviation over random draws) for each sample size. The organisers evaluate on held-out sets of both cities, which we never accessed.

Four prizes are awarded: best average F1 over the five Amsterdam models, best peak F1, best low-data F1 (n = 25) and originality. The task rules state that Amsterdam data may only be used to evaluate the method; every design decision should be made on Madrid. Section 10 discusses the one place where our development deviated from this.

Four construction periods are predicted. Classes 3 (1984–2004) and 4 (2004–2024) are shared; the boundary between classes 1 and 2 is city-specific: 1960 in Madrid and 1945 in Amsterdam. Labels are derived from **weighted_mean_year**, the area-weighted construction year of the buildings overlapping each 30 m pixel.

3 Data

Each row of the provided parquet files is one 30 m pixel in one year, with up to three Landsat observations from the summer months (Landsat 5 TM, 7 ETM+, 8 OLI, 9 OLI-2), six surface-reflectance bands (Blue, Green, Red, NIR, SWIR1, SWIR2), per-observation QA flags (QA_PIXEL, QA_RADSAT, a QA-valid flag), the scene identifier and day of year. Only pixels with at least 15% building coverage are included.

	Madrid	Amsterdam
Building pixels	76,263	25,992
Years	1984–2025 (42)	1984–2025 (42)
Class 1 / 2 boundary	1960	1945
Pixels per class (1 / 2 / 3 / 4)	13,794 / 27,152 / 18,672 / 16,645	7,568 / 8,802 / 6,745 / 2,877
Landsat path/rows	200/32, 201/32	198–199 / 23–24

Table 2. The two datasets.

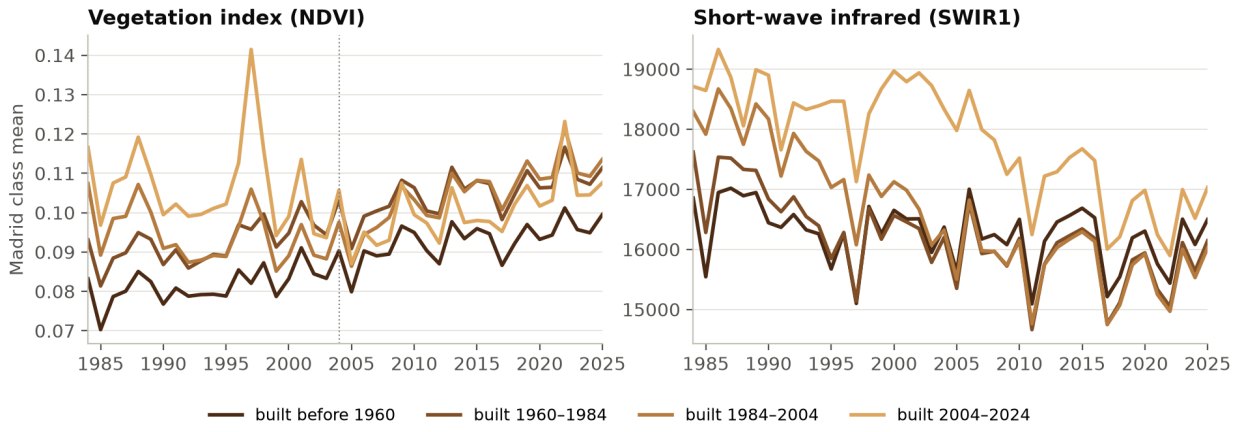


Figure 1. Average yearly vegetation index and short-wave infrared per construction period (Madrid). Newer periods start brighter in SWIR1 (bare land) and converge with the old city after construction; the timing of that convergence carries the age signal.

4 Planning and evaluation protocol

Before modelling we fixed how decisions would be made, so that results could not be steered after the fact:

- **Spatial-block cross-validation.** Madrid is cut into ~1 km blocks (33 × 33 pixels); folds are stratified by class and grouped by block (StratifiedGroupKFold). Model selection used 5 folds; from the third round on, 10 folds with an inner early-stopping split of 1/7 of the training blocks. Random splits inflate scores by 3–4 points (Figure 2).
- **Paired decisions with uncertainty.** Two configurations are compared on the same folds and seeds with the Bayesian correlated t-test (correlation $\rho = 1/k$ for k folds), with a region of practical equivalence of ±0.005 macro-F1. A change is adopted only if it is better by more than the ROPE with probability ≥ 0.95; ties go to the simpler model.
- **Rules fixed before results.** Every decision rule (ensemble size, weight averaging, shortlists, adoption) was written into the code before the run it governs.
- **Data principles.** No smoothing or deletion of observations; thresholds measured from the data rather than hard-coded; label problems handled by down-weighting, never by editing labels.
- **Leakage control.** weighted_mean_year is the label source and never an input; pixel coordinates were removed after they were shown to be a location shortcut (coordinates alone reached macro-F1 0.497 on Madrid).

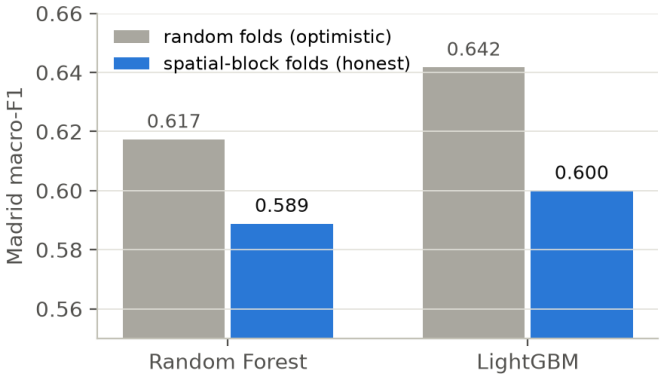


Figure 2. The same models scored with random folds and with spatial-block folds (Madrid). Random folds place test pixels next to training pixels of the same block and overstate performance.

5 Preprocessing: from raw scenes to model inputs

The preprocessing turns up to three observations per pixel and year into one value per band and year, without smoothing or deleting data. All thresholds are measured on each dataset itself.

- **Usable observation:** QA-valid (clear; no fill, cloud, shadow or dilated cloud) and the band not saturated, either by the QA_RADSAT bit of that band or because the value sits at that sensor-and-band encoding ceiling (the top value at which at least 10 observations pile up).
- **Ranking of acquisitions** (whole scenes, so a year's bands normally share a date): (1) not an anomalous acquisition — a scene whose city-wide median reflectance is a robust outlier ($|z| > 3$) among the city's scenes; (2) lower QA cloud confidence; (3) Collection-2 Tier 1 before Tier 2; (4) day of year closest to mid-summer.
- **Gaps:** years without a usable observation are linearly interpolated and flagged as unobserved.
- **Indices:** NDVI, NDBI, UI, MNDWI and BSI from the same-date bands.

- **Neighbourhood spread:** the standard deviation of each band across the 3 × 3 surrounding building pixels in that year.
- **Label weights** (training only): construction years that are heaped in the label histogram (detected with a robust z-score > 3) and labels close to a class boundary get lower weight (floor 0.2).

Group	Channels	Content
Spectral	11	6 bands + 5 indices (standardised on the training folds)
Observed mask	6	1 = real observation that year, 0 = interpolated
Day of year	1	(DOY − mid-summer) / 60 of the year's main acquisition; summer taken in the dataset's hemisphere
Sensor	5	one-hot: Landsat 4, 5, 7, 8, 9
Quality	3	share of the year's acquisitions that passed QA; anomalous acquisition; Tier 2
Neighbourhood spread	6	3 × 3 standard deviation per band
Static	2	building coverage; distance to the historic centre ÷ the city's 95th-percentile distance

Table 3. Model input per pixel: 42 years × 32 channels, plus 2 static values.

Excluded on purpose: weighted_mean_year (label source), pixel coordinates (location shortcut), the 60 per-pixel summary statistics of the starter notebook (the network computes such summaries itself), raw QA integers, and several engineered series that did not help (within-year spread, rolling standard deviation, per-pixel standardised series, change-point features; Section 6.4). A ground-up rebuild from the raw parquet files reproduced every input array bit for bit.

6 Model development, step by step

6.1 Round 1: which model family?

Thirteen model families were trained on the starter-notebook inputs. A capacity check (one batch of 128 pixels, no regularisation) confirmed each could fit the data. Under 5-fold spatial cross-validation (Figure 3), TempCNN (0.640 ± 0.008) and a hybrid TempCNN + MLP (0.641 ± 0.013) tied for first, followed by a temporal transformer (0.637). Gradient-boosted trees and random forests overfitted strongly (train − validation log-loss gaps of 0.45–0.55).

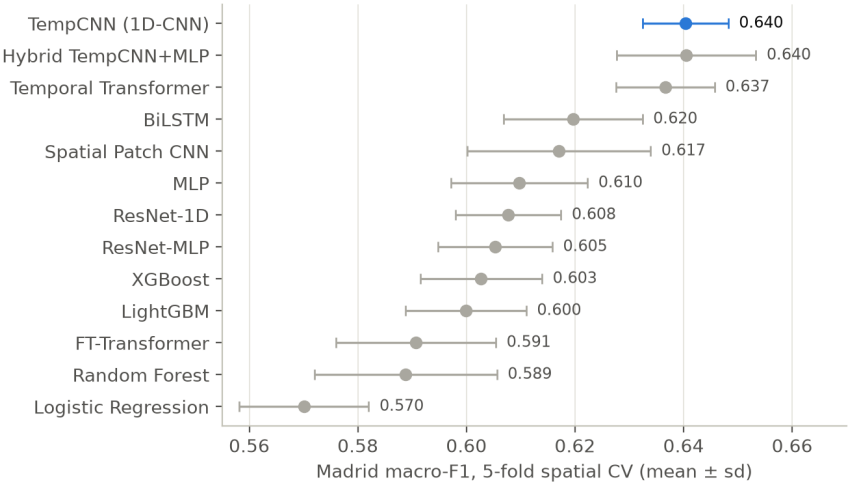


Figure 3. Round 1 ranking, Madrid macro-F1 (mean ± sd over 5 spatial folds).

6.2 Round 2: an embedding instead of a classifier

Because the goal is a *transferable* model, the networks were retrained with supervised contrastive loss (temperature 0.1, class-balanced batches) to output a 32-dimensional embedding; a pixel is classified by its nearest class prototype (mean embedding of each class). The top group was unchanged and statistically inseparable: hybrid 0.645, BiLSTM 0.639, TempCNN 0.638.

6.3 Round 3: cleaner inputs

The input rules of Section 5 replaced the starter preprocessing (which took the first QA-valid slot and dropped rows with Blue > 15,000, a rule specific to one corrupt scene). With 10-fold spatial cross-validation, TempCNN led (0.659 ± 0.015) ahead of the hybrid (0.658) and BiLSTM (0.652); TempCNN was kept as the simplest model of the leading group.

6.4 Round 4: extra feature groups

Six candidate feature groups were added one at a time. None changed Madrid cross-validation (all within ±0.003). The neighbourhood-spread channels were kept because they improved the transfer score on Amsterdam (+3.0 macro-F1 over 20 paired seeds, 95% interval [+1.5, +4.5]); per-pixel standardised series (−4) and change-point features (−6) hurt transfer. This decision used Amsterdam labels (see Section 10).

6.5 Training recipe

A fixed recipe (AdamW, learning rate 1e-3 halved on plateaus, dropout 0.3, batch 256 = 64 per class, gradient clipping 1.0, early stopping on the inner fold with a bootstrap-noise threshold) was compared with two adaptive recipes (learning-rate range test, learned temperature, dynamic class weights, gap-controlled dropout, AutoClip). The adaptive versions scored 0.651 and 0.643 against 0.661 for the fixed recipe (paired P(better) ≈ 0.01), so the fixed recipe was adopted.

6.6 Ensemble

Ten networks with different random seeds were trained per fold. Averaging their similarity scores raised macro-F1 by +0.008 over the average single network and won on 10 of 10 folds. Stochastic weight averaging over each run's plateau was slightly worse and was rejected by its pre-registered rule. The number of members is the smallest ensemble that is equivalent to all ten and better than a single network (both with P ≥ 0.95): 5.

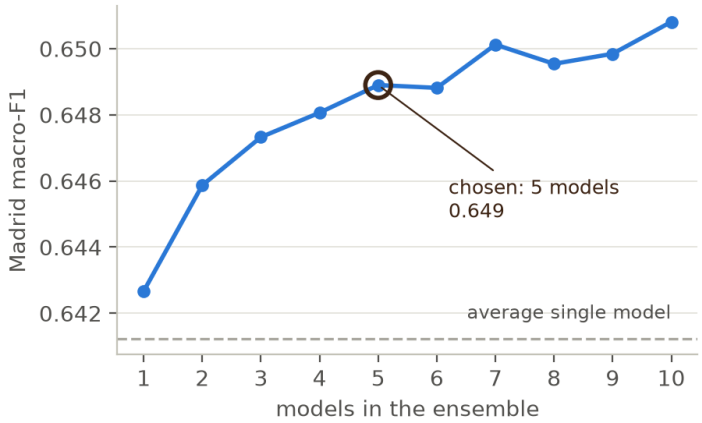


Figure 4. Madrid macro-F1 against the number of averaged networks (final build).

6.7 Architecture and loss sweep

Twenty-seven one-at-a-time changes to the architecture (filter width and dilation, number of filters, dense units, number of convolution layers, padding mode, head and embedding size) and four alternative losses (SupCon + ordinal CORAL, SupCon + year regression, proxy/normalised softmax, cross-entropy) were screened on 10 folds, about 900 networks in total. Every larger variant had its learning rate and dropout tuned on inner folds. No variant was better (best: -0.0012); none reached the shortlist (Appendix A).

6.8 Ground-up rebuild

The final backbone was rebuilt from the raw parquet files on a Kaggle T4 GPU. Every input array matched the development build bit for bit; the 10-fold assignment did not, because the platform's scikit-learn version shuffles the blocks differently. The two partitions give 0.664 ± 0.018 and 0.649 ± 0.038 for the same model: differences below about 0.01 between runs are partition noise.

7 The final backbone

Component	Specification
Input	42 years × 32 channels; 2 static values joined at the head
Convolutions	3 × [Conv1d 64 filters, width 5, zero padding → BatchNorm → ReLU → Dropout 0.3]; receptive field 13 years
Dense	flatten (64 × 42 = 2,688) → 256 units → BatchNorm → ReLU → Dropout
Head	[256 + 2 static] → 128 → ReLU → 32; output L2-normalised
Parameters	777,952 per network (88% in the dense layer)
Loss	supervised contrastive, temperature 0.1, label-reliability weights
Training	AdamW (lr 1e-3, weight decay 1e-4), batch 256 (64 per class), ≤ 40 epochs, best early-stopping epoch kept
Ensemble	5 networks (seeds 999 + 100·s), similarity scores averaged; 32 × 5-dimensional embedding
Classification	nearest class prototype (label-weighted mean embedding); kNN reported alongside

Table 4. Final backbone (stored with its normalisation statistics and metadata in backbone.pt and backbone_meta.json).

8 The transfer method: modelCreate

modelCreate(dataset, labels) prepares the new city with the same preprocessing, aligns its channels by name and its years by calendar year to the backbone, standardises with the backbone's Madrid statistics, and builds a classifier. Nothing in it refers to a particular city: the hemisphere is read from the WRS-2 rows of the dataset's own Landsat scenes (row 60 is the equator for daytime scenes), the centre is an input with a data-driven fallback (centroid of the oldest-class labels), and every distance and threshold is measured on the dataset itself.

8.1 Two modes

- **n ≤ 20 labelled pixels in the smallest class — bare backbone.** Class prototypes are the mean embeddings of the labelled pixels; each pixel takes the class of the closest prototype (ensemble-averaged cosine similarity).
- **n > 20 — full pipeline:** AdaBN (BatchNorm statistics re-estimated on the city's unlabelled pixels), few-shot fine-tuning with gradual unfreezing, prototypes and kNN, and the confidence rule.

The threshold of 20 follows from development evidence: with 5–10 labels per class, cross-validation on 20–40 labelled pixels has a standard error of about ±0.10 and occasionally selected an adaptation that cost 5–11 points; from 25 labels on, selected adaptations helped (+2.7 points on average). That analysis used Amsterdam draws.

8.2 Fine-tuning with gradual unfreezing

Stage	Trainable	Parameters	Learning rate
−1	nothing (backbone as is)	0	—
0	nothing; AdaBN statistics only	0	—
1	head	37,280	LR
2	+ dense layer and its BatchNorm	726,176	dense LR/η
3	+ last convolution block	746,848	LR/η²

Table 5. Stages. Frozen layers are excluded from the optimiser; all BatchNorm layers stay in inference mode. LR ∈ {1e-4, 3e-4, 1e-3}, η ∈ {3, 10}, and the number of epochs are chosen by spatial-block cross-validation on the

labelled pixels, with the 1-SE rule toward the simplest stage.

8.3 Spatial structure and the confidence rule

From the city's own unlabelled embeddings, the similarity of pixel pairs is measured against distance; the neighbourhood radius R is the last distance at which similarity still falls significantly (one-sided 95%), and each distance ring gets a weight proportional to its excess similarity. R was 16 pixels (480 m) in Amsterdam. For each pixel, d1 and d2 are the distances to the closest (c1) and second-closest (c2) prototypes:

- margin d2 − d1 ≥ τ → class c1, confidence **high**;
- otherwise the labelled and high-confidence pixels within R vote between c1 and c2 (ring-normalised, weighted by distance and by each voter's calibrated accuracy) → the winner, confidence **moderate-high**;
- neighbours present but none voting for c1 or c2 → c1, **low**, flagged as a conflict; no labelled neighbour → c1, **low**.

τ is the margin quantile that maximises held-out macro-F1 of this rule in the same cross-validation (fold models predict both the held-out pixels and their neighbours); if no cross-validation is possible, an Otsu split of the city's margin distribution is used.

9 Results

9.1 Madrid

10-fold spatial-block cross-validation of the final backbone: macro-F1 **0.649 ± 0.038** (mean ± sd over folds), accuracy 0.670. Per-class F1: before 1960 0.548, 1960–1984 0.676, 1984–2004 0.676, 2004–2024 0.703. Errors are concentrated between neighbouring periods; the largest is pre-1960 predicted as 1960–1984 (31%). Both periods predate the satellite record, so their histories differ only in subtle ageing signals.

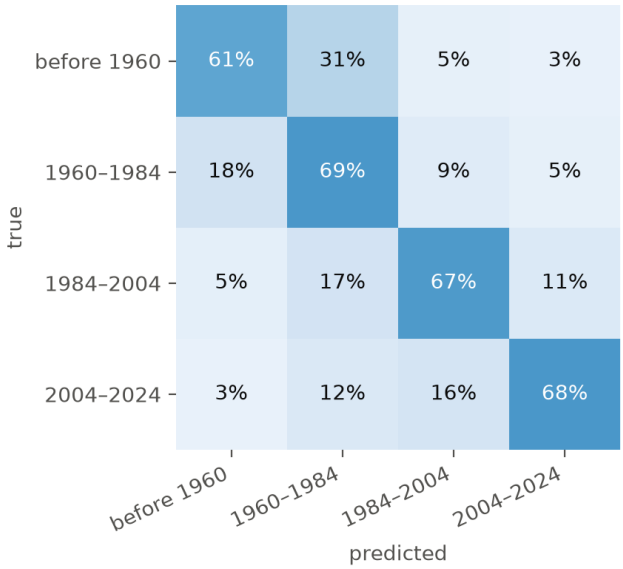


Figure 5. Madrid confusion-matrix proportions (rows = true period; all 10 folds pooled; 10-network ensemble, within 0.002 macro-F1 of the 5-network backbone).

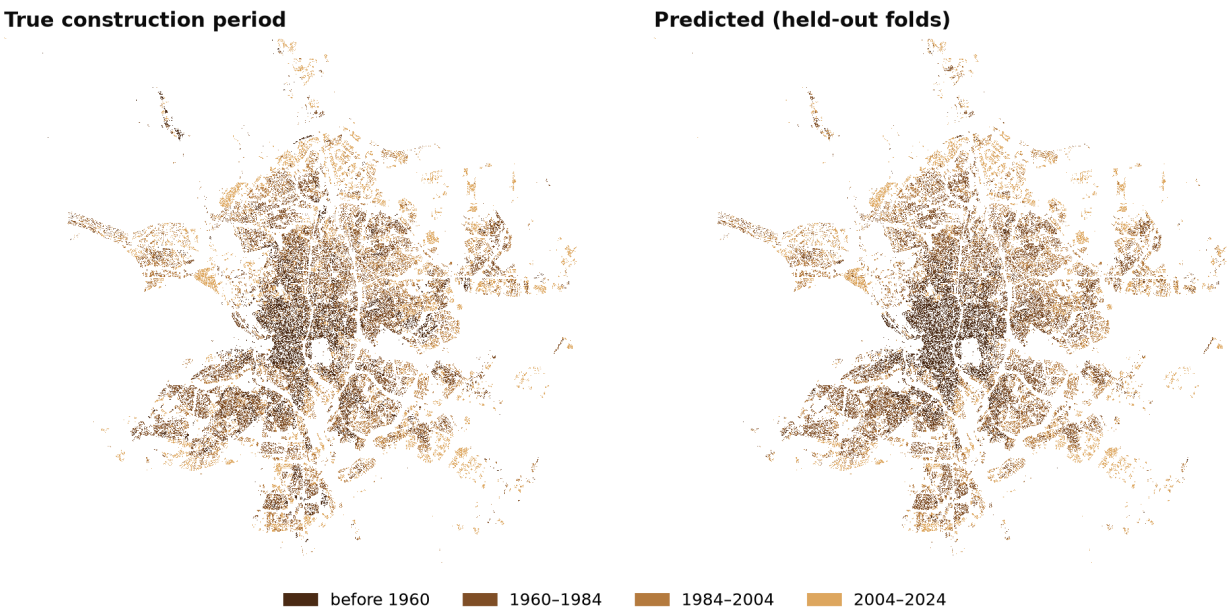


Figure 6. Madrid: true construction period and out-of-fold prediction (each block predicted by networks that never trained on it).

9.2 Amsterdam

modelCreate was run on 10 random draws of n labelled pixels per period for each n (seed 0), using the final script and the final backbone, and scored on every other labelled Amsterdam pixel.

n per period	Method	Macro-F1 mean	sd	Accuracy
5	bare backbone	0.651	0.017	0.659
25	full pipeline	0.685	0.028	0.691
50	full pipeline	0.703	0.022	0.706
100	full pipeline	0.738	0.007	0.740
200	full pipeline	0.760	0.006	0.762

Table 6. Amsterdam results. Average over n: 0.707; peak: 0.760 (n = 200); low-data (n = 25): 0.685.

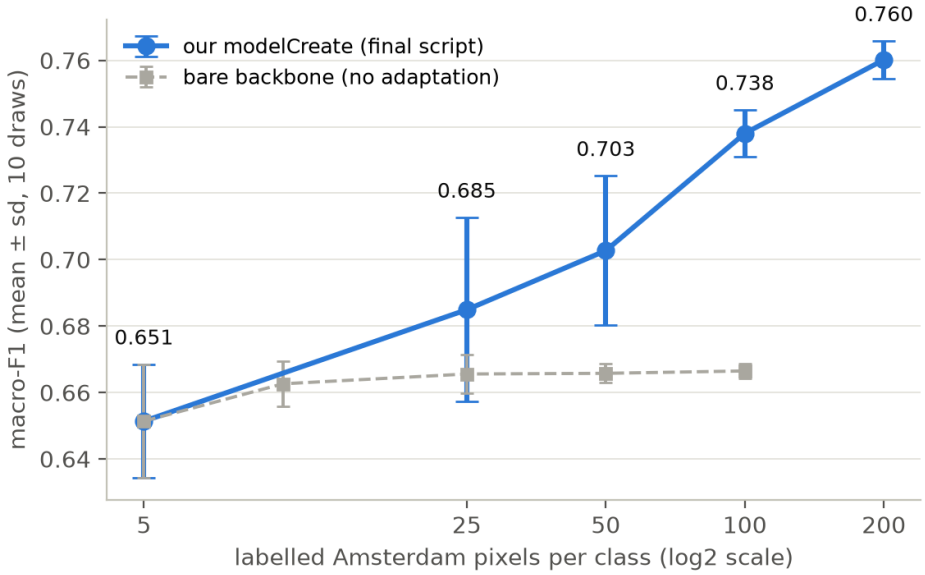


Figure 7. Amsterdam macro-F1 (mean ± sd over 10 draws) against log2 of the number of labelled pixels per period. Grey: the bare backbone with prototypes from the same draws (development evaluation, n = 5–100).

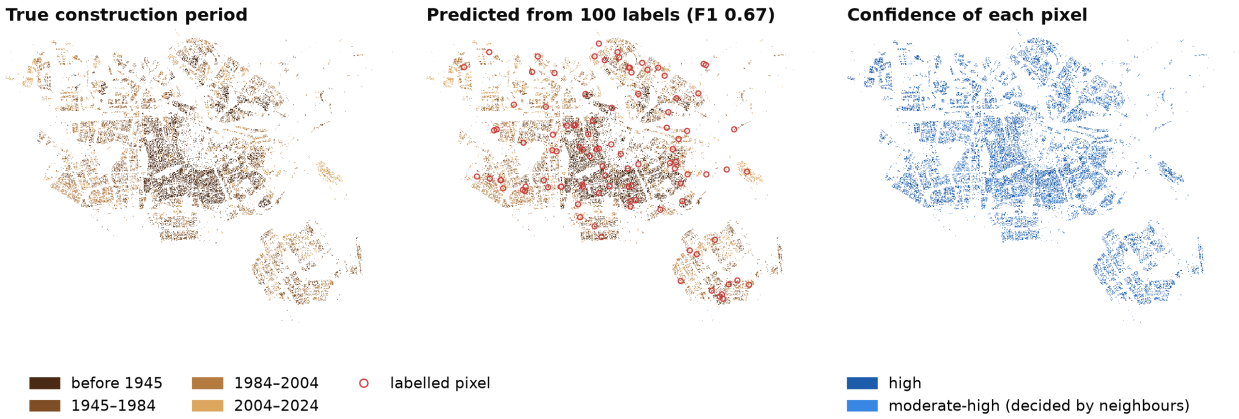


Figure 8. Amsterdam from one draw of 25 labelled pixels per period (circled): true period, prediction (macro-F1 0.67) and confidence level.

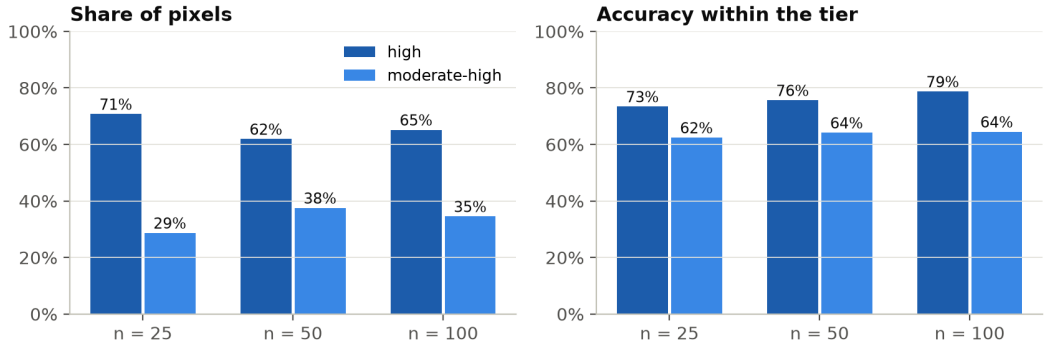


Figure 9. Confidence levels in Amsterdam (development evaluation of the same backbone and pipeline, 10 draws per n): share of pixels and accuracy within each level.

10 Discussion: what worked, what did not, limitations

Worked	Did not work
Spatial-block testing and paired decisions with uncertainty	Pixel coordinates (location shortcut; zero-shot transfer 0.05)
A model that reads the 42-year history in order (TempCNN)	Adaptive learning rate / dropout / class weights (−1 to −2 points)
Best observation per band; flags instead of deletion	Stochastic weight averaging (−0.004 to −0.005)
Supervised contrastive embedding + prototypes	Bigger, deeper, wider networks; other padding modes
Averaging 5 networks (+1 point)	Ordinal, year-regression, proxy and cross-entropy losses
AdaBN + head fine-tuning from 25 labels	Change-point and per-pixel standardised series (hurt transfer)
Neighbour-based confidence rule (+2 to +3 points)	Fine-tuning with ≤ 20 labels (selection too noisy)

Table 7. Design decisions and their outcomes.

Limitations

- **Neighbouring periods.** Most errors are one period off, especially before 1960 vs 1960–1984.
- **One feature group was selected with Amsterdam scores.** The neighbourhood-spread channels were kept after a feature ablation whose deciding evidence was transfer to Amsterdam, and the n ≤ 20 rule was motivated by Amsterdam development draws. Amsterdam never trained the backbone, but under the task rules these choices make our

Amsterdam numbers somewhat optimistic; the organisers' held-out sets give the unbiased estimate.

- **Partition noise.** Two 10-fold partitions of Madrid differ by 0.015 for the same model.
- **One training city.** Madrid's building stock and climate are not representative of every city; the bare-backbone mode relies on the embedding transferring as it is.

Next steps

- Train the backbone on several cities; validate the confidence levels on held-out data.
- Decide the n threshold and other transfer settings with a Madrid-only pseudo-city protocol (one region held out).
- Use the continuous construction year as an auxiliary signal in a way that does not cost accuracy.

11 Reproducibility

- **Single script:** building_age_pipeline.py — unpacking (parquet, folder or .zip), preprocessing, backbone loading, modelCreate and the command line; dependencies numpy, pandas, pyarrow, torch, scikit-learn (matplotlib optional).
- **Backbone:** backbone.pt with backbone_meta.json in the same folder (results/rebuild/ensemble/).
- **One model:** python building_age_pipeline.py --data city.parquet --bounds B1,B2,B3 --n 25 --seed 0 --out out/
- **Task evaluation:** add --evaluate --draws 10 (n = 5, 25, 50, 100, 200), writing mean ± sd, a table and the F1 vs log2(n) plot.
- **Compute:** development on a laptop, Kaggle T4 GPUs and an Apple M4 Pro; the final evaluation ran on the M4 Pro.
- **Seeds:** draws use NumPy generators seeded from (seed, n, draw); network seeds are fixed per fold and member.

Appendix A Architecture and loss sweep

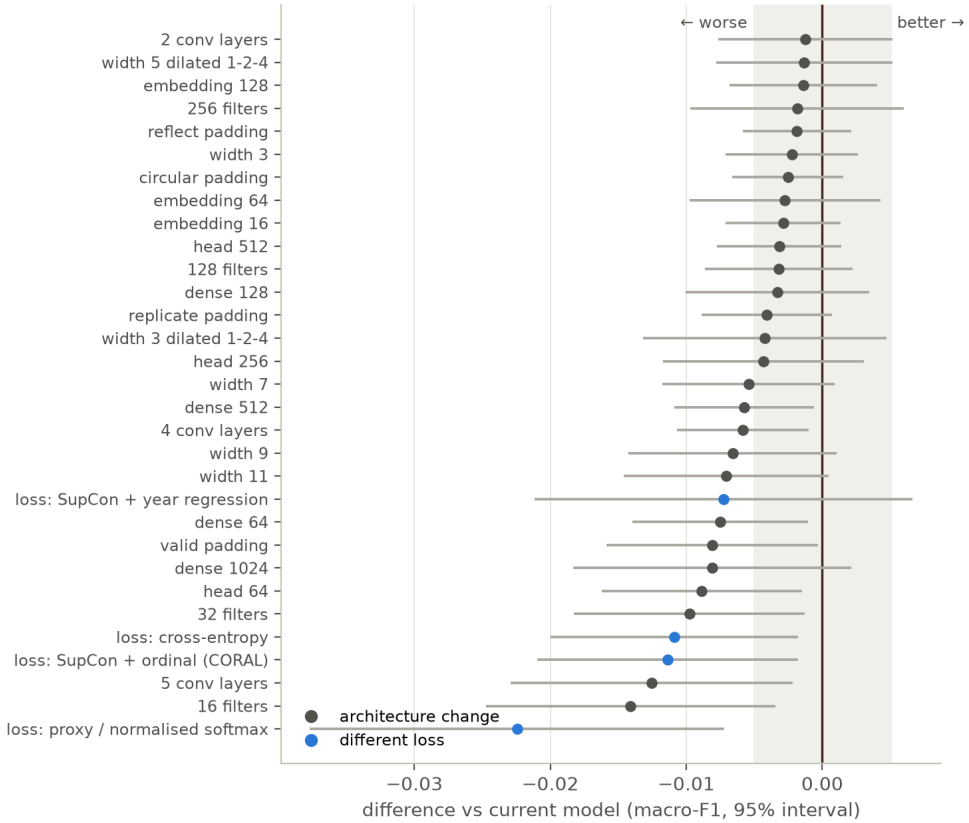


Figure A1. Paired difference to the current model (Madrid 10-fold CV, seed 0) with 95% interval; grey band: region of practical equivalence (±0.005).

Appendix B Per-fold Madrid scores

Fold	0	1	2	3	4	5	6	7	8	9
Macro-F1	0.587	0.648	0.618	0.665	0.623	0.618	0.701	0.702	0.650	0.678

Table B1. Final backbone (5 networks), 10-fold spatial-block CV (Kaggle partition).